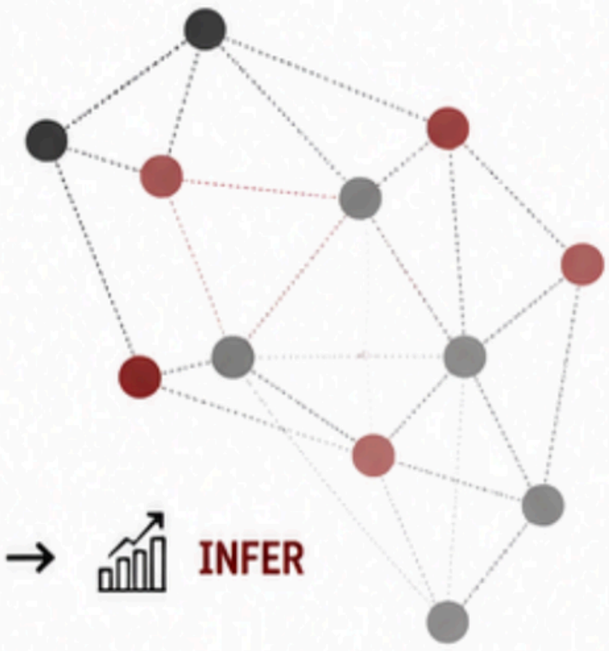
AUTONOMOUS SOCIAL REASONING AGENT

INTELLIGENCE UNDER ADVERSARIAL ARABIC CONDITIONS



جامعة عفت
EFFAT UNIVERSITY

COLLECT → NORMALIZE → REASON → SEARCH → INFER

An intelligent deliberative agent that processes Arabic social media data, applies logical reasoning and search algorithms, and generates meaningful topic and relationship insights from noisy, adversarial text.

THE PROBLEM

- Noisy Arabic social media text
- Dialect and spelling inconsistency
- Adversarial / misleading inputs
- Incomplete and uncertain data

LEADS TO

- Poor decision-making
- Low accuracy in analysis
- Missed semantic relationships



RESULTS (ACHIEVED)

38.6%	Adversarial noise reduction
92%	Inference accuracy (Forward Chaining)
0.42 sec	Average query response time
Efficient	Graph-based search (A*, BFS)

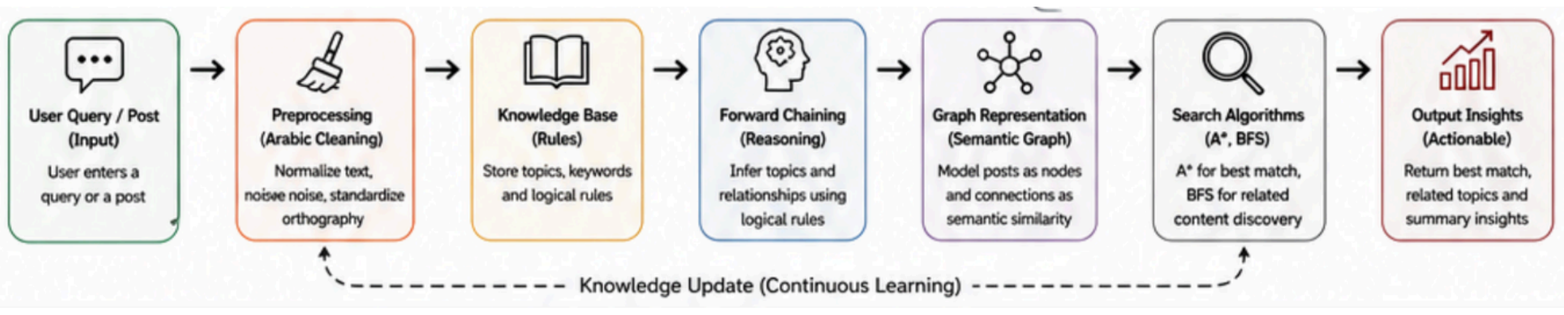
SYSTEM OVERVIEW

The system collects Arabic social data, cleans and normalizes it, reasons using a knowledge base, searches for relevant information, and generates meaningful insights.

Collect Social Posts (Social Media Data)
↓
Arabic Text Normalization (Clean & Standardize)
↓
Knowledge Base & Rules (Topics & Relationships)
↓
Reasoning Engine (Forward Chaining)
↓
Graph Search (A* Best Match, BFS Related)
↓
Insights & Related Topics (Actionable Output)

4 AGENT ARCHITECTURE

A deliberative agent that transforms noisy Arabic text into actionable insights using logical reasoning and search algorithms.



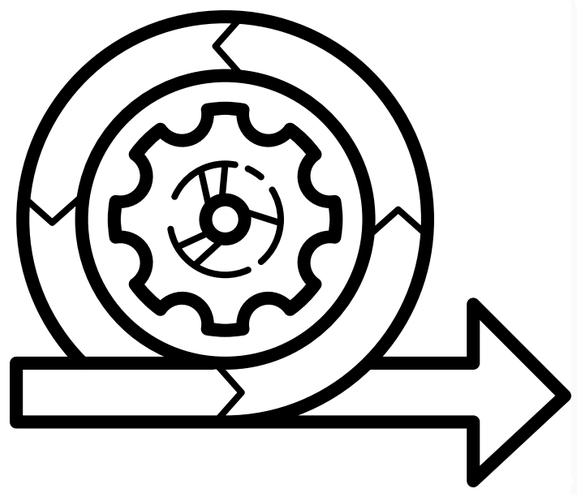
ADVERSARIAL CONDITIONS

The system operates under incomplete and noisy data conditions that challenge accurate decision making.

- **Incomplete user data** — Missing context, limited observability
- **Noisy Arabic text** — Typos, slang, repeated letters, informal writing
- **Dialect & orthographic variation** — Different dialects and spelling inconsistencies
- **Adversarial / misleading inputs** — Deceptive, sarcastic or off-topic content
- **Uncertainty in predictions** — Ambiguous intent and misleading signals

METHODOLOGY

1. **Data Collection** — Collect Arabic social media posts and user queries.
2. **Arabic Normalization** — Standardize Alif, Ya, Taa Marbuta; remove diacritics (Tashkeel) and elongation (Tatweel).
3. **Knowledge Representation** — Build a knowledge base of topics, keywords and rules.
4. **Forward Chaining Reasoning** — Infer topics and relationships from the knowledge base.
5. **Graph Modeling & Search** — Use A* (best match) and BFS (related posts) on the similarity graph.
6. **Insight Generation** — Return relevant topics, best match and related content.



7 AI PIPELINE

Data flows through structured stages to produce reliable insights.

- **Data Collection** — Collect posts from social media platforms.
- **Preprocessing** — Clean, normalize and structure the text.
- **Knowledge Representation** — Store rules, topics and keywords.
- **Reasoning Engine** — Forward chaining infers topics and relationships.
- **Search Algorithms** — A* for best match, BFS for related posts.
- **Output Insights** — Best match, related topics and network insights.

8 KEY TECHNOLOGIES

Python 3.x · Pandas · NetworkX · Graph Theory · Classical AI (Forward Chaining, A*, BFS)

9 EXPECTED OUTCOMES

- High-quality insights from noisy and adversarial data
- Faster and more accurate decision support
- Stronger data-driven strategies aligned with Saudi Vision 2030